MEHMET KEREM TURKCAN

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SUMMARY

Research scientist in physical AI with 8+ years of experience building perception, simulation, and robot learning systems for cities and medicine. Lead work on sim-to-real transfer (Constellation, Boundless), a real-time digital twin of New York City (Valdrada), humanoid navigation in Isaac Lab, VLMs and open-vocabulary 3D perception at the edge, VLA datasets for medical robotics, and LLM post-training. Published at CVPR, ICML, ACM UIST, ACM/IEEE SEC, IEEE INFOCOM, IJCV, and eLife; open-source systems with 2,200+ GitHub stars.

EXPERIENCE

◇ **Columbia University,** New York, NY — Jul 2024 – Present
Associate Research Scientist

- Lead applied AI projects across an NSF Engineering Research Center spanning 5 institutions and 30+ principal investigators, with sensing models intended to scale across 900+ NYC intersections and three urban testbeds.
- Created DART (github.com/mkturkcan/DART), a real-time open-vocabulary detector optimized with TensorRT, with public weights and 790+ GitHub stars; led UrbanOmniDetect (github.com/mkturkcan/urbanomnidetect), an oral paper at the CVPR 2026 DriveX Workshop.
- Built Valdrada (github.com/mkturkcan/valdrada), a real-time NYC digital twin (258 GitHub stars in six weeks), and G1 Patrol (github.com/mkturkcan/g1_patrol), Unitree G1 navigation in crowds trained in Isaac Lab.
- Lead surgical world model research for medical robotics; mentored student first authors at ICML 2025, IEEE INFOCOM 2026, and USENIX VehicleSec 2026; co-authored the ACM/IEEE SEC 2025 Best Paper.
- Secured two NVIDIA research grants, with Z. Kostic (2025) and A. Smyth (2026), and EmpireAI project support; lead HPC, data, deployment, and public release workflows; main engineering instructor for CS3 RET in 2024–2026.

◇ **Columbia University,** New York, NY — Nov 2022 – Jul 2024
Postdoctoral Research Scientist

- Built real-time urban digital twins integrating detection, tracking, trajectory prediction, simulation, and safety alerts; developed Unreal Engine 5 synthetic data and high-altitude detection benchmarks.
- Built healthcare AI pipelines with Northwell Health for surgical phase recognition and automated endoscopy skill scoring; mentored 11 Master's and three high school researchers.

◇ **Columbia University,** New York, NY — Jan 2023 – Dec 2023
Lecturer in Electrical Engineering

- Designed and taught graduate courses in neural networks and advanced deep learning to 123 students.

◇ **Columbia University,** New York, NY — Jan 2017 – Oct 2022
Doctoral Researcher

- Co-created FlyBrainLab, combining ontology retrieval, OrientDB graph querying, GPU simulation, and 3D visualization.

EDUCATION

Columbia University – Ph.D., Electrical Engineering, GPA 4.10 — 2022
Columbia University – M.S., Computer Science, GPA 3.97 — 2016
Istanbul Technical University – B.S., Electronics and Communication Engineering, GPA 3.75 — 2015

SELECTED PUBLICATIONS

◇ **Constellation: Benchmarking High-Altitude Object Detection for an Urban Intersection** — IJCV 2026
◇ **Vulnerable-User-Aware Urban Transportation Digital Twin: Methods and Applications** — IEEE T-ITS 2026
◇ **Digital Eyes on the Road: Street Cameras to Verify Traffic Integrity at City Scale** — USENIX VehicleSec 2026
◇ **Car-Following-Informed Gaussian Processes for Vehicle Trajectory Prediction** — TRR 2026
◇ **Zero-Shot Image Tagging at Traffic Intersections via SAM-Guided VLMs** — ICMLA 2026
◇ **Calibration-Free View-Agnostic Monocular 3D Object Detection for Urban Scenes** — CVPRW 2026, Oral
◇ **Worst-Case Attacks in Reactive Edge-Cloud Systems** — IEEE INFOCOM 2026
◇ **Real-Time Video Analytics for Urban Safety** — ACM/IEEE SEC 2025, Best Paper
◇ **Adaptive Data Collection for Robust Learning Across Multiple Distributions** — ICML 2025
◇ **StreetNav: Street-Camera Navigation for Blind Pedestrians** — ACM UIST 2024
◇ **FlyBrainLab: Accelerating Discovery of the Functional Logic of the Drosophila Brain** — eLife 2021

TECHNICAL SKILLS AND SERVICE

◇ **AI and vision:** PyTorch, JAX, TensorFlow, Hugging Face, detection, tracking, 3D perception, VLMs, VLAs, LLM post-training, RAG, agents, diffusion and world models
◇ **Deployment and systems:** Python, C/C++, CUDA, TypeScript, TensorRT, ONNX, LiteRT, WebGPU, Jetson, HPC, OrientDB, Docker, CI/CD
◇ **Simulation and robotics:** NVIDIA Isaac Lab and Isaac Sim, Newton (MuJoCo-Warp), Unreal Engine 5, CARLA, OpenUSD, ROS 2; Unitree G1 humanoid robot learning and sim-to-real transfer
◇ **Reviewer:** CVPR 2026, ICML 2026, ICMLA 2026, WACV 2025, MobiCom 2024, IEEE/ACM Transactions on Networking, IEEE TVLSI, and Sensors (MDPI)